

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 3 Summer 2025 (Supply.)

Course : Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTMC303 - THERMODYNAMICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 A transformer while working is an example of
A) Open system B) close system c) can't be predicted
d) isolated system
- 2 A cylindrical drum which has the diameter of 1.5 m, a length of 4.2 m and contains 3000 kg of liquid, the density would be
A) 400 kg/m³ B) 400.2 kg/m³ C) 300.2 kg/m³
D) 350 kg/m³
- 3 The absolute pressure of the systolic blood pressure measured as 120 mm of Hg gauge pressure of the healthy person is
A) 880 mm of Hg B) 640 mm of Hg C) 760 mm of Hg
- 4 The temperature rises of a system is 10 °C, the temperature rise in K as
(a) 18 K (b) 10 K (c) 15 K (d) none of the above
- 5 The COP of a refrigerator with the refrigerator temperature as -10 °C if the ambient temperature is 30 °C
(a) 5.5 (b) 5 (c) 3.1 (d) 6.6
- 6 The area under a reversible process on the P-v diagram represent
A) heat transfer
B) work transfer
C) entropy transfer
D) all of these
- Degree of superheating of the superheated steam at the temperature 200 °C and atmospheric pressure is
A) 100 B) 45 C) 65
D) can't calculate
- 8 **Slop of the melting curve on the P-T diagram of the water is**
A) positive
B) negative
C) zero
D) can't say
- 9 The Molecular weight of a gas having $c_p = 1.968$ and $c_v = 1.50$ kJ/kg K is
A) 16 kg/kg mol B) 18 kg/kg mol C) 17 kg/kg mol D) can't be predicted

- 10 Entropy generation of a real process always
 A) positive B) negative C) Zero D) all of the above
- 11 Characteristic gas constant of the nitrogen gas N_2 is
 A) 0.262 kJ/kg K B) 0.297 kJ/kg K C) 28 kJ/kg K D) 0.287 kJ/kg K
- 12 Dryness fraction of saturated water is

- A. Less than 1
 B. Equal to 1
 C. Greater than 1
 D. 0

Q2. Solve the following.

Define the following with suitable example : Open system, close system and isolated system

6

A mixture of gas expands from 0.03 m^3 to 0.06 m^3 at a constant pressure of 1 MPa and absorbs 84 kJ of heat during the process. Find the change in internal energy of the mixture.

Q3. Solve the following.

- A) State the first law of thermodynamics and explain the concept of PMM-I with suitable illustration. 6
- B) A blower handle 1 kg/s of air at 200°C and consumes a power of 15 kW. The inlet and outlet velocities of air are 100 m/s and 150 m/s respectively. Find the exit air temperature, assuming adiabatic conditions 6

Q4. Solve Any Two of the following.

Show that COP of a heat pump is greater than COP of refrigerator by 1 operate at same temperature limits. 6

Show that entropy of the Universe always increases 6

An inventor claims that his heat engine has the following specifications, power 50 kW, fuel burnt per hour 3 kg, heating value of the fuel 75000 kJ/kg, temperature limits 627°C and 27°C , Justify his claim. 6

Q5. Solve Any Two of the following.

- A) Starting from relation of specific heats c_p and c_v of the ideal gas with temperature show that $c_p - c_v = R$ 6
- B) Obtain the value of Universal gas constant from first principal? 6
- C) A certain gas has $c_p = 1.968$ and $c_v = 1.507$ kJ/kg K. A constant volume chamber of 0.3 m^3 capacity contains 2 kg of the gas at 5°C . Heat is transferred to the gas until the temperature is 100°C . Find the work done, the heat transferred and the changes in internal energy. 6

Q6. Solve Any Two of the following.

Define the Dryness fraction and obtain the expression on the enthalpy of the wet steam in terms of respective enthalpies and dryness fraction. 6

Find the saturation temperature, the change in entropy and enthalpy of evaporation during evaporation of water at 1 MPa. 6

Define the following terms: Saturation temperature, superheated steam, Enthalpy of evaporation 6

*** End ***